

EMCH 308 Semester Project Rubric – Fall 2026

Criterion	Points	Scoring
Meets weight specification	15	Full points for beams less than or equal to 1 lb. 1 point off for beams 1.001–1.05 lbs, 2 points off for beams 1.051–1.1 lbs, etc.
Meets factor of safety (FoS) specification	15	Full points for beams with at least a 1.25 FoS for a load of 5,000 lb. 1 point off for FoS 1.2–1.249, 2 points off for FoS 1.15–1.199, etc.
Meets cost of manufacturing specification	15	Full points for beams with manufacturing cost of \$600 or less. 1 point off for cost \$600.01–\$650.00, 2 points off for cost \$650.01–\$700.00, etc.
Communication	15	Full points if all communication guidelines followed. Grading to be updated.
Comparison between finite element prediction and experimental measurement of the midline deflection at the center span for applied loads up to 6,250 lbf.	10	Full points for a plot showing simulated and experimentally measured curves with less than 5% deviation in slope. 1 point off for every additional 5% error in the slope.
.CAE file and .ODB files of the final simulation are submitted and contain sound modeling approaches	15	1, 2, or 3 points will be deducted per misconfigured input depending on how consequential the error is.
A short report comparing simulation and experimental results that briefly explains the modeling approach and any deviations between the predicted and experimental testing results.	15	Full points if: 1. The modeling approach and assumptions are clearly stated. 2. Rationale for the elements selected and checks for relevant numerical artifacts are included. 3. Potential effects of element size are discussed. 4. A general comparison of predicted to experimentally measured data is present. 5. Any discrepancies between simulation and experiment are explained.